



PLASTICS

Plastics are amazing materials. They don't rot like wood or rust like some metals, and they are light and easy to shape. Plastic pens, shoes, and even surfboards are all made from oil or coal. Chemicals are taken from these fossil fuels and turned into small, white pellets.

These are then melted and blown to form bags or rolled flat to make floor tiles. Buckets, bowls, and boxes are usually shaped by being injected into molding machines.



Bowled out

Plastic bowls are made by injecting melted plastic pellets into the space between two halves of a steel mold. The plastic cools inside the mold, and sets into a bowl shape.

Cold water cools down the plastic after it has been molded.



Coloring and plastic pellets are poured in.

Heaters help to melt the pellets.

New bowl

Mold

A screw pushes the squashed, hot pellets into the mold.

Melted plastic

Each bowl needs this amount of coloring.

Two handfuls of polypropylene plastic pellets make each bowl.

Temperatures of 540°F (280°C) are needed to melt the pellets and make them flow into this mold.



Get set or go?

Some plastics are like bread! Once they have been "baked" they

cannot be reheated and made into new shapes. A mug made of melamine will not change shape when hot drinks are poured into it. Polystyrene and polythene are more like chocolate—they can be melted again and again. Each time the mixture cools, it sets into the shape of the mold it has been poured into.



The two halves of the steel mold lock tightly together.

